

MID TERM EXAMINATIONB.TECH PROGRAMMES (UNDER THE AEGIS OF USICT) — 3rd SEMESTER, NOVEMBER 2022**Paper Code: ES-201****Subject: Computational Methods****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Marks	CO														
1(a)	Find a root of the equation $x^3 - 4x - 9 = 0$ using Bisection method.	[2.5]	CO1														
1(b)	Define the Absolute, Relative and Truncation Errors.	[2.5]	CO1														
1(c)	Evaluate $\Delta^3[(1 - 6x)(6 - x)(5 - 3x)]$.	[2.5]	CO2														
1(d)	Use Trapezoidal rule to estimate the integral $\int_0^1 [dx / (1 + x)]$.	[2.5]	CO2														
2(a)	Perform four iterations of the Newton-Raphson method to obtain the approximate value of $(17)^{1/3}$ starting with the initial approximation $x_0 = 2$.	[5]	CO1														
2(b)	Minimize $f(x) = x^2 + 54/x$ in interval $(0, 5)$ using Fibonacci search method.	[5]	CO1														
3(a)	For the data: <table><tr><td>X</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>F(x)</td><td>15</td><td>5</td><td>1</td><td>3</td><td>11</td><td>25</td></tr></table> Construct the Newton's forward difference table and find $f(-1.5)$.	X	-2	-1	0	1	2	3	F(x)	15	5	1	3	11	25	[5]	CO2
X	-2	-1	0	1	2	3											
F(x)	15	5	1	3	11	25											
3(b)	Find the value of $f(9)$ from the given table: <table><tr><td>X</td><td>5</td><td>7</td><td>11</td><td>13</td><td>17</td></tr><tr><td>F(X)</td><td>150</td><td>392</td><td>1492</td><td>2366</td><td>5202</td></tr></table>	X	5	7	11	13	17	F(X)	150	392	1492	2366	5202	[5]	CO2		
X	5	7	11	13	17												
F(X)	150	392	1492	2366	5202												
4(a)	Round off the numbers 86.5250 and 37.46235 to four significant figures and compute Absolute error, Relative Error and Percentage error in each case.	[5]	CO1														
4(b)	Find the polynomial of lowest possible degree which assumes the values 3, 12, 15, -21 when x has the values 3, 2, 1, -1 respectively.	[5]	CO2														

****** End of Question Paper ******